

Neural Network-Based Control of Forced-Convection and Thermoelectric Coolers

Mahsa Akhsham^{1b}, Mohammad Javad Dousti^{1b}, and Saeed Safari^{1b}

Abstract—Thermoelectric cooling technology holds great promise as a spot-cooling solution for the VLSI circuits. Typically, it is combined with a forced convection cooler, such as a fan to achieve effective thermal management. The optimal driving current for the thermoelectric coolers and fan rotation speed play a crucial role in minimizing cooling power consumption while ensuring safe operating temperatures for the VLSI chip dies. Various software-based optimization frameworks have been proposed to calculate the near-optimal values in real time. Nonetheless, these frameworks suffer from several limitations, including slow runtime, high power consumption, costly optimization software license, and the need for a separate computer to execute the optimization software. This article presents a hardware-friendly neural network (NN)-based optimizer that can learn from the functionality of a software optimizer. In order to train the NN model, a self-supervised learning strategy is employed to generate the desired labeled data. Simulation results demonstrate that the proposed model successfully meets the thermal constraints across all the tested benchmarks, with an overall system cooling power consumption increase of only 3.6% compared to a software optimizer. Furthermore, the hardware implementation of the proposed model proves to be more than five orders of magnitude faster than the software-based optimizer, consuming a negligible amount of power while occupying a small die area.

Index Terms—Cooling, dynamic thermal management (DTM), forced-convection cooling, low-power design, neural network (NN), self-supervised learning (SSL), thermoelectric coolers (TECs).

I. INTRODUCTION

RECENT technological advances and shrinking transistor sizes have led to increased chip power density. High-power density adversely affects the performance of different components and accelerates the aging process. As a result, thermal issue is one of the main barriers to further improve the speed of the electronic chips and must be controlled by an appropriate thermal management system.

Different solutions to manage and alleviate thermal issues have been proposed during the past few decades. The effectiveness and advantages of these cooling methods vary depending on the specific properties of a target system and power densities it needs to manage. For instance, microchannel

liquid cooling is ideal for dissipating background heat on the large chips and 3-D-stacked architectures. Nevertheless, the temperature of the fluid increases as it traverses the channels, leading to a decrease in heat removal capability at locations distant from the liquid inlet [1]. Thermoelectric coolers (TECs) excel at handling high power densities in compact areas, but they consume significantly more power when applied to the cool larger areas. Consequently, an approach that combines the strengths of various cooling methods, known as hybrid cooling, has the potential to deliver considerably higher efficiency.

Recent hybrid cooling designs include combining TECs with microchannel liquid cooling [2], [3], or with heatsink and fan cooling [4], [5], [6], [7]. Existing works on using TECs and liquid cooling mostly focus on localized use of TECs for high-intensity hot spots and targets much higher heat fluxes [2], [3]. However, liquid cooling setups are generally more complex than the air cooling solutions, requiring additional components, such as a pump, tubing, and a liquid cooling block. Also, this cooling system is usually more expensive due to the additional components and complexity and may require more maintenance, such as checking for leaks and refilling coolant [4]. As a result, in this work, we target a combination of TECs, a fan, and a heatsink to dispose the heat generated from the surface of a chip. The setup of this combination is relatively straightforward to manufacture and deploy. In addition, it is more cost effective and the maintenance requirements are typically lower than that of the liquid cooling systems.

Simultaneous control of the fan speed and TECs' driving current such that the entire system meets its thermal and power constraints is challenging. If the TECs' driving current is not correctly set and the rotation speed of the fan is not adjusted appropriately, power efficiency and cooling performance can be negatively affected. High TECs' driving current with low fan speed causes the heat pumped from the surface of a chip to be trapped above TECs. On the other hand, low TECs' driving current with high fan speed does not cool hot spots. Finally, TECs are active cooling devices which consume power and produce heat through the Joule heating. Therefore, TECs' driving current and the fan rotation speed are the two correlated variables and should be optimally set to minimize the chip temperature or the total cooling power. Another crucial factor to consider is the chip's leakage power, which exhibits an exponential dependency on the temperature despite advancements in new FinFET technology [8], [9], [10]. Furthermore, the value of leakage power is significant compared to the total

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chip power consumption and can range from 25% to 44% of it [11] in room temperature. Considering leakage power increases a thermal model's accuracy but transforms it from a linear system to a nonconvex one [12], which makes it challenging to find the optimum fan speed and TECs' driving current.

Dousti and Pedram [5] suggested an optimization framework called optimization of forced-convection and TECs (OFTECs) to minimize the cooling-related power consumption while adhering to a maximum die temperature constraint. This framework uses a software-based global optimizer to determine near-optimum fan speed and TECs' driving current in order to minimize the maximum chip temperature. Using a software-based global optimizer limits the usability of OFTEC in several ways. First and foremost, executing a software optimizer to calculate desired controlling values is inefficient and requires an additional computer to control the cooling system. Second, high-quality software optimizers are expensive and usually cost several thousand dollars [13]. Third challenge emerges due to the comparatively slower performance of software optimizers in contrast to their hardware equivalents. Consequently, utilizing a software optimizer typically results in higher energy consumption as hardware optimizers accomplish tasks much faster under identical conditions. Last but not least, hardware optimizers consume considerably less power than their software counterparts.

All these limitations encourage using a hardware-based optimizer to replace the software solution suggested by OFTEC. Note that, the optimization problem of interest (OFTEC) is nonconvex and requires a careful design of a hardware solution. In this article, we propose a neural network (NN)-based optimizer called NN-OFTEC and provide a hardware implementation of it. NNs are a suitable tool due to their ability to capture nonlinearity of complex objective functions [14]. Moreover, there have been many recent advancements in hardware implementation of them [15], [16], [17], [18], [19]. This facilitates effective implementation of a given NN on the hardware. Toward this end, we use a multilayer, fully connected architecture, which is customized for our problem and uses a self-supervised learning (SSL) strategy [20] to prepare the data for training the network. Next, we design a 5 nm ASIC for our proposed network and show that it meets power and thermal objectives.

We show that the additional die size for the NN-OFTEC is very small, i.e., 0.045 mm^2 for the Alpha 21 264 processor. Besides, use of a hardware optimizer eliminates the high power consumption of running a software optimizer on a PC while it consumes negligible power, i.e., $23.9 \mu\text{W}$. Moreover, the runtime of NN-OFTEC is more than the five orders of magnitude faster than that of OFTEC.

The main contributions of this article are summarized below.

- 1) Designed NN-OFTEC, which is a neural-network optimizer to solve a nonconvex optimization problem in order to control a cooling system comprised of TECs, a fan, and a heatsink. This system is on a par with a software optimizer in terms of control quality.
- 2) Trained NN-OFTEC using the SSL strategy.

- 3) Implemented NN-OFTEC on ASIC and performed a thorough formality equivalence check between the software and hardware implementations.
- 4) Decreased energy consumption by improving the response time of OFTEC by five orders of magnitude while consuming a negligible amount of power.

The design of NN-OFTEC will be open-sourced upon the acceptance of this article. This liberates interested designers from licensing issues involved in using software optimizers.

The remainder of this article is organized as follows. Section II reviews two main subjects of interest: 1) principles of thermoelectric cooling and 2) the combination of constrained mathematical optimization and NNs. Section III presents thermal models and corresponding optimization problem formulations used in this article. Section IV explains how to design the corresponding OFTEC NN. Section V presents the experimental results. Section VI concludes this article.

II. PREVIOUS WORK

A. Thermoelectric Coolers

Many studies have explored the use of TECs for cooling high-performance multiprocessor chips [3], [5], [6], [7], [21], [22], [23], [24], [25], [26], [27], [28]. Biswas et al. [21] used TECs to cool down microprocessors in a data center and reduced the total cooling cost while maintaining high reliability. This article mainly focuses on the steady-state analysis of TECs and uses a constant coefficient of performance (COP) for modeling them. As a result, the fan speed and power consumption are assumed to be constant. Hence, the method is not very accurate. Alexandrov et al. [22] showed the significance of the transient behavior of TECs in cooling a VLSI die. It presents two simple controllers: 1) a threshold-based controller, which turns ON or OFF TECs when the temperature goes above or below a specific temperature and 2) a maximum cooling-based controller, which uses the hysteresis effect to decrease the number of ON or OFF transitions in TECs. TECs are supplied with a constant current to effect a state change in both the controllers. Dousti and Pedram [24] incorporate the leakage power into the TEC thermal model and suggests a new COP formulation for the entire cooling package. Then, the optimum TEC current for achieving the minimum temperature or the maximum COP for the system is found through the simulations for the whole chip package. Papers mentioned above do not account for any additional cooling method, and TECs alone cannot sufficiently cool scorching spots and avoid the thermal runaway situation. So, another cooling technique should be considered to enhance the effect of TECs.

Thermoelectric conversion technology represents an alternative application of TECs, extensively integrated into numerous engineering processes, such as providing thermoelectric energy to the microelectronic systems [26], [27], [28]. This technology directly transforms a fraction of thermal waste energy into electrical energy using a thermoelectric generator (TEG). This generated electrical energy is adequate for low-power electrical loads, such as low-power sensors or electronic

chips. Similar to TECs, TEGs offers notable benefits, including a cost-effective system, high service life, noise-free operation, and environmental sustainability. Nevertheless, TEGs are potentially suitable for several applications where efficiency is not the primary concern [28]. Furthermore, our contribution centers on the cooling scenarios, whereas those work emphasize on the energy generation.

Kaplan et al. [3] developed a compact hybrid cooling model, which is able to account for the thermal behavior of liquid microchannels together with TECs. They also proposed LoCool, a cooling power optimization method for the systems that adopt hybrid cooling. LoCool optimizes hybrid systems combining microchannel-based liquid cooling and TEC cooling for the hotspot mitigation in a localized manner. Even though they target very dense hotspots in a localized manner, they did not take into account the processor's temperature-dependent leakage power in their model. In the context of runtime optimization algorithms, they generate a lookup table of the optimum TEC current values corresponding to various hotspot heat flux levels. It is imperative to note that the feasibility of this approach is limited due to the impracticality associated with employing an individual current source for each TEC. Such a strategy proves economically burdensome and consumes substantial power. Also note that simply employing a lookup table was feasible due to the fact that highly nonlinear leakage power was not taken into account.

Some other works suggested a convex function to formulate the multiprocessor temperature without using TECs and try to solve an optimization problem, such as [23] and [25]. In these solutions, however, TECs are not used and also solving a convex optimization problem is time consuming and is not recommended for prompt decisions during runtime of multiprocessor systems.

Dousti and Pedram [5] in their next work, present a nonconvex thermal model for chip cooling comprised of TECs and a fan. A formulation for minimizing the cooling power subject to the system's thermal and physical constraints was set up in which optimization variables were the TECs' driving current and fan rotation speed. Since, the objective function considers leakage power, the suggested formula has a non-convex form, and a software-based optimization framework called OFTEC was developed to address this issue. Results showed that OFTEC is a suitable framework for finding the near-optimal solutions and meets all the system's thermal and physical constraints. Unfortunately, it relies on the complex optimization software to find solutions. To avoid the high complexity of OFTEC, TECfan [7] employs a greedy algorithm to individually turn ON and OFF TECs while employing a constant high current for TECs. They have also considered independently modifying fan speed and chip's voltage and frequency (DVFS). TECfan does not consider the overhead of selectively enabling TECs [6], while also tune fan speed and turn on TECs independently.

In order to consider enjoying the benefit of OFTEC, i.e., variable current control of TECs, while addressing its high overhead, one needs to implement it in hardware. In this article, we choose NN to learn the behavior of OFTEC and

present a hardware implementation for the proposed NN. The next section reviews the literature in this regard.

B. Constrained Optimization Using Neural Networks

Current research areas in the intersection of constrained optimization (CO) and NNs can be categorized into two main directions. The first one focuses on using NN to aid the performance of the CO problem solvers. The second one, which is of our interest, focuses on integrating optimization solvers into deep learning architectures.

Many methods focused on developing NN architectures to predict fast and approximate solutions to the predefined CO problems, such as Hopfield networks [29], Lagrangian duality [29], [30], and differentiable optimization layers [31]. However, these approaches suffer from various weaknesses. For instance, the Hopfield network has notable sensitivity to parameter initialization and hyperparameters and focuses on discrete optimization problems not continuous ones [29]. Alternatively, the existing methods for differentiable optimization layers [31] are not flexible enough in order to be used for different problems, such as nonconvex ones.

In the context of multicore thermal management, Amos and Kolter [32] used deep Q-learning (DQL) to maximize the power efficiency of heterogeneous multiprocessors. Nevertheless, authors evaluate their work through well-known benchmarks and do not address challenges of new application models and requirements. Moreover, the runtime design parameters are limited to the three levels of operating frequency and determining the number of active processors. Gupta et al. [33] addressed dynamic energy optimization of multiprocessor systems under performance constraints through NNs. For this purpose, their work uses DVFS, discarding other runtime parameters. Similarly, Moghaddam [34] and Ul Islam et al. [35] proposed an NN-based framework for energy optimization of embedded systems through DVFS, while neglecting other system-level design parameters. Likewise, Zhang et al. [36] took advantage of QL to independently compute the optimal voltage and frequency of the cores in multicore platforms for power and performance optimization.

Otoom et al. [37] proposed a core temperature estimation method using NNs for the preemptive fan control. Authors consider long intervals (in the order of minutes) for workload variations such that each core temperature reaches its steady state. Realistically, in multitasking environments, workload varies much faster [39], [40]. Terraneo et al. [39] proposed a transient fan model, enabling adaptive fan speed control simulation for efficient dynamic thermal management (DTM). Besides, they propose a reinforcement learning-based DTM policy which leverages fan speed, DVFS, and dynamic core assignment. This policy jointly maximizes the performance, while minimizes the fan power subject to the thermal constraints. This article does not consider TECs, while our objective is to jointly control TECs and a fan.

As can be seen, NNs are useful tools for replacing the traditional optimization solvers, but they need to be designed

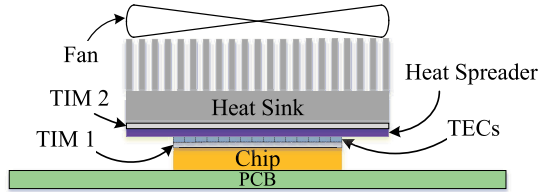


Fig. 1. Chip assembly with its cooling solution [5].

properly based on the problem of interest. In the next section, we discuss the thermal model used by OFTEC and its problem formulation. Next, we propose an NN architecture to replace OFTEC.

III. MODELING AND FORMULATION

Reference [5] uses the duality between the electrical and thermal phenomena to make a circuit model of a cooling package like the one presented in Fig. 1.

In this model, each physical component is decomposed into several subcomponents. Increasing the number of these subcomponents increases the accuracy of the model; however, it also increases the complexity of the electrical circuit model; thus making the analysis slow.

The processor package contains eight layers as shown in Fig. 1: 1) PCB; 2) chip; 3) TIM1; 4)TEC; 5) heat spreader; 6) TIM2; 7) heat sink; and 8) fan.

Layers 1, 3, 5, and 6 are denoted by L_{conduct} in this work. Subcomponents in L_{conduct} only conduct heat (i.e., they do not generate or absorb heat) and are modeled as resistances. Layer 2 is called L_{chip} , which generates heat, and the power consumption in this layer has two parts: 1) *dynamic* and 2) *leakage power*. Dynamic power is independent of the temperature and is not affected by the cooling solution. On the other hand, the leakage power depends exponentially on the temperature. To calculate the leakage power, Iranfar et al. [40] suggested only using the linear term of the Taylor series in the expansion of the leakage power equation. This linear estimation for the subcomponent i can be written as follows:

$$p_{i,\text{leakage}} = a(T_i - T_{\text{ref}}) + b \quad (1)$$

where a and b are the Taylor expansion coefficients, T_i is the temperature of the subcomponent i , and T_{ref} is the reference temperature around which the Taylor series is expanded. This temperature is usually set as the average temperature of the chip or a particular functional unit inside the chip to increase the accuracy of the estimation and consequently speed up the aforesaid iterative method.

The TEC layer shows three behaviors: 1) heat absorption; 2) heat rejection; and 3) heat generation and is broken into three sublayers: 1) $L_{\text{TEC,Abs}}$; 2) $L_{\text{TEC,Rej}}$; and 3) $L_{\text{TEC,Gen}}$. The power of each subcomponent i in the sublayers above can be calculated as follow:

$$p_i = -\alpha I_{\text{TEC}} T_i, \quad i \in L_{\text{TEC,Abs}} \quad (2)$$

$$p_i = \alpha I_{\text{TEC}} T_i, \quad i \in L_{\text{TEC,Rej}} \quad (3)$$

$$p_i = R_{\text{TEC}} I_{\text{TEC}}^2 + \alpha \Delta T I_{\text{TEC}}, \quad i \in L_{\text{TEC,Gen}} \quad (4)$$

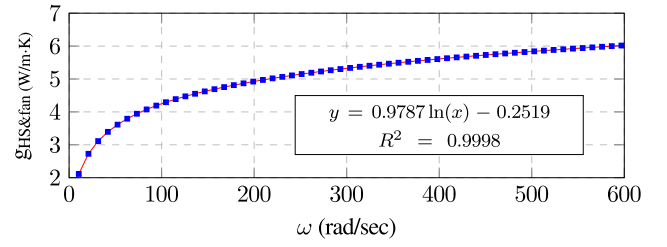


Fig. 2. Dependence of the thermal conductance of combined heat sink and fan on the fan rotation speed [5].

where ΔT is the temperature difference between the upper (the hot side) and the lower (the cold side) subcomponents and (4) defines the power consumption of one TEC unit. Other equations listed above, i.e., the power of absorption and rejection, are for modeling and do not contribute to the power consumption of a TEC. Finally, layers 7 and 8 are categorized as $L_{\text{HS\&fan}}$. For the laminar airflow, the power consumption of a fan as a function of its rotation speed, ω may be estimated as

$$P_{\text{fan}} = c \cdot \omega^3 \quad (5)$$

where c is a constant which depends on the air viscous friction, air density, and the radius of the fan blades [41]. The thermal conductance of the heat sink depends on the airflow. A fan can change the airflow. Therefore, the collective thermal conductance of the heat sink and the fan can be written as a function of ω . Using the calculation methodology employed in HotSpot v7.0 [42], Fig. 2 shows the dependence of the thermal conductance of the heat sink and the fan ($g_{\text{HS\&fan}}$) on the fan rotation speed. By performing curve fitting on Fig. 2, $g_{\text{HS\&fan}}$ can be estimated as

$$g_{\text{HS\&fan}} = p \cdot \ln(q \cdot \omega) + r, \quad \omega \gg 1 \text{ rad s}^{-1} \quad (6)$$

where p and r are the fitting parameters, which depend on the material and physical properties of the heat sink, the fan, and air (such as air density and thermal conductivity). Parameter q is added to make the logarithm value dimensionless so that both the sides have the same unit dimension.

In this article, such as [5], this value is simply considered as 1 s and p and r are adjusted accordingly. For small values of ω , $g_{\text{HS\&fan}}$ can be estimated as the thermal conductance of heat sink (g_{HS}), which is constant under steady conditions of ambient air. We aim to minimize the cooling power consumption of the entire chip package subject to the system temperature constraint. Since, the leakage power is a function of temperature, which is affected by the cooling efficiency, it is also included in the objective function. This problem can thus be formulated as shown in Optimization 1. The optimization variables in this formulation are ω and I_{TEC}

Optimization 1: Cooling power minimization subject to the thermal and the physical constraints

$$\underset{\omega, I_{\text{TEC}}}{\text{minimize}} \quad \mathcal{P} := P_{\text{leakage}} + P_{\text{TEC,N}} + P_{\text{fan}} \quad (7)$$

where

$$P_{\text{leakage}} = \sum_{i \in L_{\text{chip}}} p_{i,\text{leakage}} \quad (8)$$

$$P_{\text{TEC},N} = \sum_{i \in L_{\text{TEC},\text{Gen}}} p_i \quad (9)$$

$$P_{\text{fan}} = c \cdot \omega^3$$

subject to

$$\mathbf{G}(\omega)\vec{T} = \vec{P}(\omega, I_{\text{TEC}}) \quad (10)$$

$$T_i < T_{\text{max}}, \quad i \in L_{\text{chip}} \quad (11)$$

$$0 \leq \omega \leq \omega_{\text{max}} \quad (12)$$

$$0 \leq I_{\text{TEC}} \leq I_{\text{TEC},\text{max}}. \quad (13)$$

Equations (7)–(9) define the terms in the objective function, (7) defines the leakage power as the sum of the leakage power of the subcomponents ($p_{i,\text{leakage}}$) in the chip layer, (8) expresses the power consumption of all the N TECs as the sum over the power consumption of all the TECs in the TEC generation sublayer, which was given in (4), and (9), like (5), defines the fan power consumption.

In the following, constraints are presented, (10) is a system of equations derived from the KCL equations for all the nodes in the equivalent electrical circuit. The total current (dual of power in the thermal model) that leaves a node (the left-hand side) is equal to the current (power) that enters a node (the right-hand side). Matrix $\mathbf{G}$ is defined as follows:

$$\mathbf{G}(\omega) = \begin{bmatrix} \sum g_{1,i} & -g_{1,2} & \cdots & -g_{1,n} \\ -g_{2,1} & \sum g_{2,i} & \cdots & -g_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ -g_{n,1} & -g_{n,2} & \cdots & \sum g_{n,i} \end{bmatrix} \quad (14)$$

where $g_{i,j} = g_{j,i}$ is the thermal conductance between the subcomponents i and j . These values are constant except the thermal conductance between the heat sink/fan and the ambient that is equal to $g_{\text{HS}\&\text{fan}}$. This value is a function of ω as described in the previous section; hence, the matrix $\mathbf{G}$ is a function of ω . $\vec{T}$ keeps the temperature of all the subcomponents in the thermal model where the temperature of the subcomponent i is denoted by T_i . $\vec{P}$ contains the power consumption of all the subcomponents in all the layers in the thermal model, where the power consumption of the subcomponent i is denoted by p_i . The previous section presents the definition of p_i values for each layer's subcomponents. Note that, the chip layer's dynamic power consumption of the subcomponents is considered an input to the problem. $\vec{P}$ is a function of both the optimization variables, i.e., I_{TEC} and ω . Here, it can be seen that using a linear estimation for the leakage power instead of a constant value does not add any computational complexity to (10) because it is already a system of linear equations regarding T_i values. As it is explained earlier, this estimation speeds the leakage power calculation.

Constraint (11) ensures that the temperature of all the subcomponents in the chip layer remains below a certain threshold (T_{max}), and (12) and (13) enforce the physical limitations. More precisely, (12) sets an upper bound (ω_{max}) and a lower bound (zero) for the rotational speed of the fan, whereas (13) imposes an upper bound ($I_{\text{TEC},\text{max}}$) and a lower bound (zero) on the driving current of TECs.

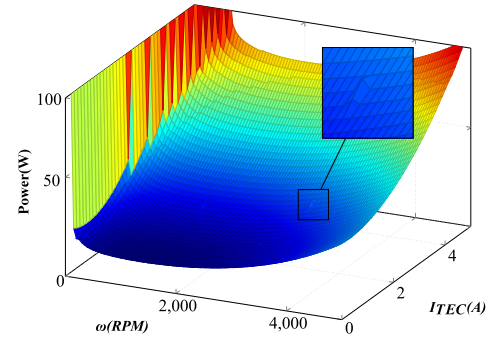


Fig. 3. Objective function of Optimization 1 for Basicmath benchmark [5].

If the TEC current exceeds this threshold, the TEC will be damaged.

Fig. 3 shows the objective function of Optimization 1 (i.e., $\mathcal{P}$) drawn with respect to optimization variables while running the Basicmath benchmark [43]. As can be seen, Optimization 1 is not convex. Hence, this problem is classified as a constrained nonlinear program (CNLP). In order to solve this problem, OFTEC uses the MATLAB's active-set SQP method [44] to produce near-optimal solutions for the TECs' driving current and fan rotation speed.

Employing a software-based optimizer, the usability of OFTEC is limited in several ways. In the first place, calculating desired controlling values using a software optimizer is inefficient and requires an additional computer. Second, MATLAB and its global optimization toolbox cost over 3000 U.S. dollars [13]. The third problem arises from the higher energy consumption of a software optimizer compared to the hardware optimizers, which complete tasks much more quickly and consume a negligible amount of power, resulting in lower energy consumption. As a result, we propose using a hardware-based optimizer to replace the software solution suggested by OFTEC. Due to the nonconvexity of our problem, NNs are a proper solution. Therefore, we design an NN-based OFTEC framework called NN-OFTEC, which provides a hardware implementation for the NN of interest.

IV. NN-OFTEC FRAMEWORK WORKFLOW

NN-OFTEC framework workflow is depicted in Fig. 4. In the remainder of this section, first the design, training, and ASIC implementation of the NN-OFTEC model are explained. After that, the data generation process to train and validate the model is explained.

A. Neural Network Design and ASIC Implementation

The NN architecture should have a small hardware footprint while inferring accurate results and satisfying all the constraints listed in (10) and (13). For this purpose, we employ a fully connected network with a standard ReLU as the activation function of hidden layers and a limited-ReLU in the range of $[0, 1]$ as an activation function of the output layer. Considering that the training data are applied to the network in a normalized manner and limited-ReLU is used in the final layer, (12) and (13) are implicitly satisfied during

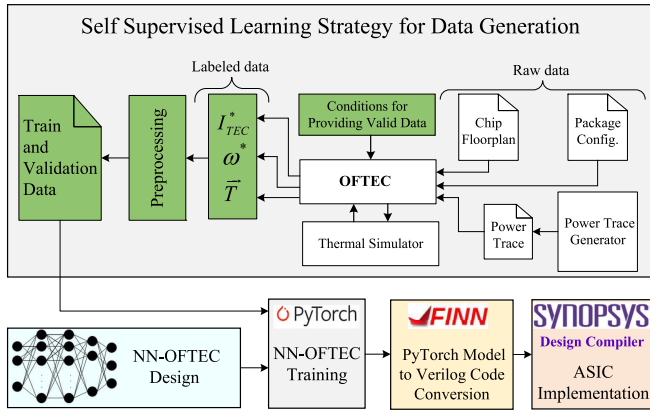


Fig. 4. NN-OFTEC framework workflow.

the inference. To capture (10) and (11), a dummy node is added to the output layer in order to predict the maximum die temperature. In this manner, the effect of temperature constraint is applied in the backpropagation process and the network is trained according to the thermal constraints. Note that, this technique eliminates the need for using the complex power and temperature equations (as a loss or activation function), which facilitates the learning process and simplifies the network design.

The network depth and the number of neurons are determined based on the complexity and number of optimization variables. If an objective function is linear, one layer can classify the data, but more hidden layers are needed if it has quadratic or higher complexity [45]. Based on the nonconvex nature of Optimization 1, a network with three hidden layers is employed. The number of neurons in the first and final layers is determined based on the size of the input features and optimization variables. Considering that the temperature vector serves as the input and ω^* and I_{TEC}^* serve as outputs, the number of neurons in the first layer equals the size of the temperature vector, and there are two neurons plus a dummy node in the output layer. The determination of the number of neurons in every hidden layer and other hyperparameters of NN-OFTEC is accomplished through the design exploration on the validation dataset. The neurons in both the hidden layers and the output layer are contingent upon the optimization formula with only the number of neurons in the first layer being modified by the input feature size and the count of hotspots. Consequently, our model, formulated for OFTEC and Optimization 1 formula is scalable and extends to various chips. The sole adjustment required pertains to the number of neurons in the input layer, a parameter dependent on the number of hotspots identified by the temperature sensor.

Ultimately, the configuration of NN-OFTEC as a fully connected network can be succinctly represented by specifying the number of nodes in each layer and denoting it as

$$\{\text{\# of components/hotspots}, l_1, l_2, l_3, 3\} \quad (15)$$

where l_1 , l_2 , and l_3 represent the number of neurons in hidden layers one to three, respectively.

The next step is to train the NN-OFTEC model using the training data (which is explained in the next section.) We

use the sum of mean-squared error (MSE) of outputs as the loss function and *RMSPprop* [46] with the stochastic gradient descent as its optimizer. During training, we ensure that the MSE loss reduces both for the training and validation sets to avoid overfitting for our NN model. As shown in Fig. 4, training is carried out in PyTorch [47], and then, the model is implemented in Verilog using the Xilinx FINN framework [48]. Finally, the model is synthesized via *Synopsys design compiler* and verified via *Synopsys formality* [49].

B. Self-Supervised Data Generation

Train and Validation Data: In order to train and validate an NN, it is necessary to have proper data. Inspired by the self-distillation SSL technique [20], we prepare labeled data as follows. First, numerous power traces (i.e., $\vec{P}$) are generated. Using these power traces, chip floorplan, and package configuration, OFTEC solves Optimization 1 and finds ω^* and I_{TEC}^* along with $\bar{T}$. These three sets of data form the labeled data. In other words, we use the temperature vector as the input features, and ω^* and I_{TEC}^* as the data labels. Note that, the data should be carefully generated for the network convergence. Invalid data points are filtered before creating data partitions. More concretely, the valid data should meet all of the following conditions.

- 1) There should be no temperature runaway in the chip.
- 2) Calculated temperature values should not exceed $T_{\max}$.
- 3) Each component has a specific power density and cannot withstand more. So, the generated power traces must be within the acceptable range of component's power density. This condition is especially important for smaller and more commonly used parts, such as arithmetic logic units.

Accordingly, some of the generated data are omitted in this step. Furthermore, we utilize learning curve analysis to determine the minimum amount of data which prevents underfitting or overfitting in our network. Finally, features and labels are min-max normalized in the $[0, 1]$ range using the following equation:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}. \quad (16)$$

The resultant normalized data were split into the training and validation datasets with a ratio of 80% to 20%.

Test Data: In order to generate the real-world test sets, we choose well-known benchmarks or real power trace files for each chip floorplan. Next, OFTEC is used to find near-optimum TECs' driving current, fan speed, and the corresponding $\bar{T}$ to generate a test dataset. It is crucial to highlight that the test set belongs to a completely different distribution and NN-OFTEC remains unaware of this test set's specifics during the training phase.

V. EXPERIMENTAL RESULT

A. Simulation Setup

Experiments target an Alpha 21264 processor [50], an Intel Xeon X5550 processor [6], [51], an Intel Core-i7 6950X

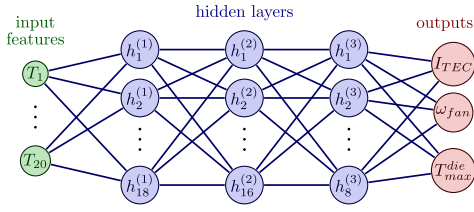


Fig. 5. NN-OFTEC architecture for Alpha 21264.

processor taken from the PACT [52] repository,¹ and a 3-D floorplan taken from HotSpot v7.0 example files.² Alpha 21264 is a single core processor with up to 20 hot spots which demonstrates relatively lower power consumption compared to the other chips. On the other hand, the Intel Xeon X5550 and Intel Core-i7 6950X are both the multicore processors but differ in terms of hot spot density and distribution. The Xeon X5550 has four cores, eight threads, and up to 58 hot spots, whereas the Core-i7 6950X boasts ten cores, 20 threads, and have up to 72 hot spots. Additionally, the fourth floorplan includes a 3-D chip with up to 21 hot spots, necessitating an unique analytical approach. Train, validation, and test data are generated as noted in the previous section. More specifically, the number of data points meeting all the conditions specified in the previous section for the Alpha 21264, Xeon X5550, Core-i7 6950X, and 3-D chip are 1500, 4500, 5500, and 2000, respectively. As explained earlier, the training and validation datasets are formed by splitting these datasets in an 80% to 20% ratio. In order to generate the real-world test sets for the Alpha 21264 processor, we use the eight MiBench benchmarks [43]. PTscalar [53] is used as the performance and power simulator to generate the dynamic power trace for each benchmark. The test data for the Xeon X5550 processor is acquired from the PARSEC benchmark [54]. Regarding the Core-i7 6950X and 3-D chip floorplans, power trace files recommended in their respective repositories are utilized as the test data, and they are denoted by PT_# and PT'_#. $\omega_{\max}$, $I_{\text{TEC},\max}$, and $T_{\max}$ are set to 524 rad/s (=5000 RPM), 5 A, and 90 °C (363 K), respectively. Moreover, the ambient temperature of the chip is assumed as 45 °C (318 K.) Based on the Alpha 21264 processor architecture and after performing various design explorations, NN-OFTEC architecture has the configuration of {20, 18, 16, 8, 3} and is depicted in Fig. 5. The Xeon X5550, Core-i7 6950X, and the 3-D chip exhibit the same hidden/output layers and neurons with the Alpha 21264 processor, but varying counts of input neurons, corresponding to the number of components delineated in their respective floorplan files. Accordingly, the NN-OFTEC architecture for the Xeon X5550, Core-i7 6950X, and 3-D chip has 58, 72, and 21 neurons in the input layer, respectively. Simulation results are performed on a system with an Intel Core-i7 9700k 3.60 GHz CPU and 128 GB of memory.

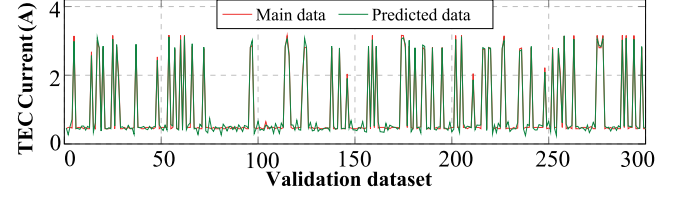
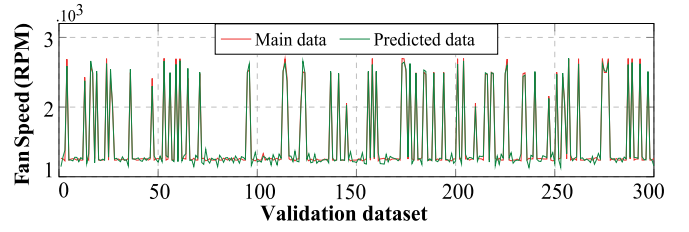


Fig. 6. NN-OFTEC validation test results. (a) Fan speed output. (b) TEC current output for Alpha 21264 processor.

B. Simulation Results

Fig. 6(a) and (b) show predictions of the trained model for the validation dataset for the Alpha 21264 processor. The results show that NN-OFTEC closely tracks OFTEC after training, indicating that the selected training dataset is adequate for training this network. Besides, NN-OFTEC results meet the die temperature constraint.

Fig. 7 compares the chip cooling power consumption and maximum die temperature for OFTEC, LoCool, and the proposed NN-OFTEC in four different chips. For every chip, TECs are selectively deployed only on the top of the hotspots. For instance, cache memory does not exhibit any hotspot and hence was not covered with any TEC. For the 3-D chip, the methodology recommended in [55] is employed. As explained in the referenced work, for each hotspot, a configuration involving two TECs is adopted with one is placed at the bottom of the chip and another is located at the top in order to pump heat for the bottom to the top. To ensure a fair comparison, the LoCool's proposed method was implemented based on our experiment setup (including the floorplan and hot spot patterns derived from the benchmarks.) We matched the number of TECs used in each processor for both the OFTEC and LoCool. Subsequently, we calculated the TEC current value and liquid-microchannel velocity of each hybrid cooling block according to the LoCool's algorithm [3].

Fig. 7(a) shows OFTEC along with NN-OFTEC and LoCool do not violate the $T_{\max}$, while running CPU-intensive workloads. Fig. 7(b) shows the chip cooling power consumption. In some benchmarks, NN-OFTEC consumes slightly more power than OFTEC. NN-OFTEC outperforms OFTEC in the Quicksort benchmark of the Alpha 21264 processor because it allows a higher maximum temperature while respecting $T_{\max}$ and in the other three benchmarks, results are almost identical. As the power consumption of the LoCool system, including the power of a TEC and liquid-microchannels, scales linearly with the number of TECs, we observed that its cooling power exceeds that of OFTEC and NN-OFTEC in Alpha 21264, Xeon X5550, and Core-i7 6950X processors, where up to 72 TECs were employed. In the

¹<https://github.com/peaclab/PACT>

²<https://github.com/uva-hotspot/HotSpot>

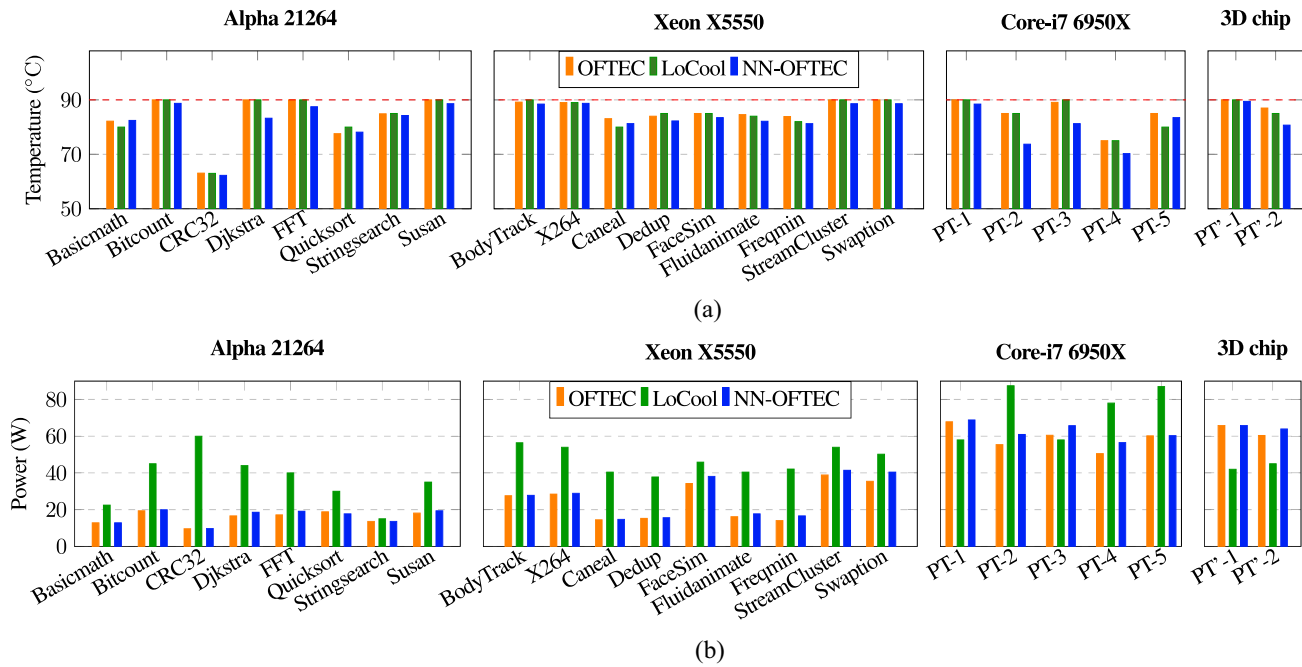


Fig. 7. (a) Maximum chip temperature. (b) Total cooling power for different benchmarks and floorplans.

TABLE I
RUNTIME COMPARISON OF OFTEC AND SOFTWARE IMPLEMENTATION
OF NN-OFTEC FOR MiBENCH BENCHMARK

Benchmark	OFTec Runtime (ms)	NN-OFTEC Runtime (ms)
Basicmath	131	0.09
BitCount	65	0.11
CRC32	291	0.09
Dijkstra	273	0.08
FFT	185	0.19
Quicksort	76	0.08
Stringsearch	161	0.07
Susan	65	0.11

TABLE II
NN-OFTEC ASIC SYNTHESIS RESULTS

Chip	Area (mm ²)	Inference Time (μs)	Power (μW)
Alpha 21264	0.045	3.00	23.90
Xeon X5550	0.050	3.32	26.60
Core-i7 6950X	0.071	4.76	38.03
3D chip	0.046	3.04	24.30

Based on the presented outcomes, it can be asserted that NNs represent a fast and effective approach for acquiring proficiency in optimization problems.

suggested 3-D structures for OFTEC, two sets of TECs are employed for cooling. However this redundancy is absent in LoCool's design due to its utilization of liquid-microchannel cooling. As a result, the power consumption in LoCool's configuration is lower than that of OFTEC and NN-OFTEC.

Compared to OFTEC, NN-OFTEC predictions cause 3.6% more power consumption compared to those of OFTEC which is not substantial. In three of the selected processors, NN-OFTEC also outperforms LoCool due to the higher number of hot spots/TECs. However, in the 3-D chip, LoCool outperforms OFTEC because of its hybrid cooling design. Overall, these results demonstrate the capability of NN-OFTEC.

As mentioned before, software optimizers are relatively slow and solve the optimization problem of interest in hundreds of milliseconds. The response time of each MiBench benchmark which is used for the Alpha 21264 processor is shown in Table I. As shown, the response time of the software-based NN-OFTEC is three orders of magnitude faster than OFTEC.

C. ASIC Synthesis

As a final step of our work, we performed the ASIC synthesis of NN-OFTEC to streamline the integration of NN-OFTEC within the chip package. We used Synopsys design compiler and a 5 nm-FinFET standard cell library with a supply voltage of 0.45 V [56], and a clock period of 20 ns to synthesize the Verilog code of our design. The results of area, inference time, and power consumption of NN-OFTEC for four chips are presented in Table II. As shown, the inference time of NN-OFTEC in each processor is in order of microseconds, which signifies a substantial improvement in speed, as it is five orders of magnitude faster than the results generated by OFTEC (see Table I.)

The reported area in the first column of Table II is taken from the design compiler report. For instance, the total area of NN-OFTEC for the Alpha 21264 processor is 45 487 μm², of which the combinational area is 38 241 μm², and the noncombinational area is 7246 μm². The original Alpha 21264 processor was realized through the utilization of a 0.35 μm CMOS fabrication process, incorporating a notable count of

15 million transistors and occupying an area of approximately 3.1 cm^2 [50]. Assuming ideal scaling in VLSI design, when the technology reduces by a factor of $1/S$, the corresponding area decreases by a factor of $1/S^2$ [57]. Consequently, considering a technology size of 5 nm, the Alpha 21264 processor would possess an area approximately equal to 6.32 mm^2 . Hence, NN-OFTEC for the Alpha 21264 occupies only 0.7% of the total die area. In the case of other chips, considering the dimensions specified in their floorplan files, NN-OFTEC occupies less than 1% of the overall die area.

The power consumption of NN-OFTEC, encompassing both the dynamic and leakage power and reported in design compiler simulation is illustrated in the third column of Table II. Notably, the power consumption associated with the NN-OFTEC design is minimal and can be considered negligible.

The postsynthesis circuit functionality was also validated using *Synopsys formality* [49], and results indicate that the synthesized circuit and reference model are equivalent. These results demonstrate the feasibility of implementing NN-OFTEC using 5 nm technology and 0.45 V supply voltage.

VI. CONCLUSION

A hybrid cooling assembly for the chips, consisting of TECs and a fan, presents a challenge in determining the appropriate TEC driving current and fan rotation speed due to the nonconvex nature of the optimization problem. Previously, a software-based optimization framework, called OFTEC was proposed to solve this optimization problem. However, the utilization of a software-based optimizer is accompanied by certain limitations, including elevated power consumption resulting from the requirement of an additional computer running complex optimization software, slow prediction speed which causes more energy consumption, and the high cost associated with procuring an optimization software license. To overcome these limitations, this article proposes a hardware-based optimizer named NN-OFTEC, which utilizes a multilayer, fully connected NN trained using an SSL generative strategy. Results demonstrate that NN-OFTEC successfully meets thermal constraints across all the tested benchmarks. Moreover, the hardware implementation of NN-OFTEC exhibit significantly improved performance compared to OFTEC, being five orders of magnitude faster. The adoption of a hardware optimizer also eliminates the excessive power consumption associated with running a software optimizer on a separate computer. Last but not least, for an Alpha 21 264 processor, NN-OFTEC consumes a mere 23.9 μW and has an area overhead of only 0.7%.

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